

FA23-BCE-113
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MSI-Lab # 5

Task 1: Assembly language program

Write an assembly language program to

- a) Display “?”
- b) Read two decimal digits between 0 to 4
- c) Display the result as per the following format

Sample execution:

?24

The sum of 2 and 4 is 6

Hint: INT 21h always read input values as ASCII characters

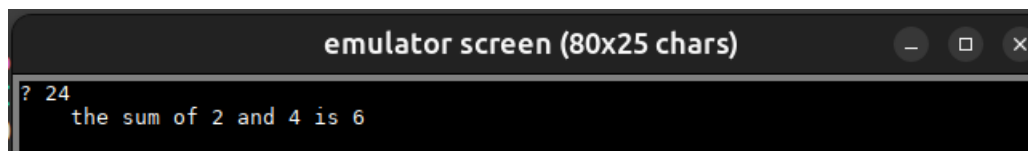
```
.MODEL SMALL
.STACK 100h
.DATA
; Variables and constants are defined here
disp DB '? $'
msg1 DB 0ah , 'the sum of $ '
msg2 DB " and $"
msg3 DB " is $ "
.CODE
MAIN PROC
; Instructions will be written here
MOV CX, @data
MOV DS , CX
; disp ?
MOV AH , 09h
LEA DX , disp
INT 21h
; > read operand 1
MOV AH, 01h      ; Read char without echo | 2 for echo
INT 21h          ; al = read data
MOV CL , AL
;sub cl , 30h     ; cl = operand 1
; > read operand 2
MOV AH, 01h      ; Read char without echo | 2 for echo
INT 21h          ; al = read data
MOV CH , AL
;sub ch , 30h     ; ch = opernad 2
; > disp msg1
MOV AH , 09h
LEA DX , msg1
INT 21h
; > disp operand 1
MOV DL, CL      ; copy cl > DL
MOV AH, 02h      ; Display function
INT 21h          ; displays the character in DL
; > disp msg2
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    MOV AH , 09h
    LEA DX , msg2
    INT 21h
; > disp operand 2
    MOV DL, CH      ; copy ch > DL
    MOV AH, 02h     ; Display function
    INT 21h        ; displays the character in DL
; > disp msg3
    MOV AH , 09h
    LEA DX , msg3
    INT 21h
; converting to assci
    SUB CL , 30h    ; cl = operand 1
    SUB CH , 30h    ; ch = opernad 2

; > caluclating sum
    ADD CH ,CL ; ch = sum result
    ADD CH , 30h
; > disp sum
    MOV DL, CH      ; copy ch > DL
    MOV AH, 02h     ; Display function
    INT 21h
MAIN ENDP
; Procedures other than MAIN will be written here
END MAIN

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Task 2: Write an assembly language program to convert the case of a string

Write a complete assembly program that converts a user-input lower-case string into corresponding upper-case string.

Sample execution:

Input: hello world

Output: HELLO WORLD

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.model small
.stack 100h
.data
    PROMPT DB 'Enter letters in lowercase: $'
    BUFFER DB 20 , ? , 20 DUP('$')
                ; max length 20
                ; actual length typed
                ; input storage, with $ to terminate
    MSG2 DB 0Dh,0Ah, 'uppercase : $'
.code
main PROC
    MOV AX, @data

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MOV DS, AX

; show prompt ah = 9
LEA DX, PROMPT
MOV AH, 09h
INT 21h

; read string ah = 10
LEA DX, BUFFER
MOV AH, 0Ah
INT 21h

; message2
LEA DX, MSG2
MOV AH, 09h
INT 21h

; get number of characters entered
MOV CL, [BUFFER + 1] ; size of char string
MOV SI, OFFSET BUFFER + 2 ; start of actual input

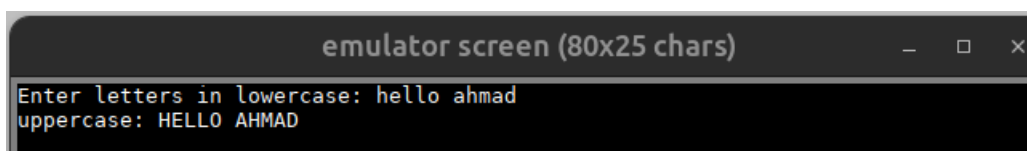
convert_loop:
MOV AL, [SI]
CMP AL, 0Dh ; stop at Enter (carriage return)
JE done
SUB AL, 20h ; adding 32(20h) for converting to
uppercase
MOV [SI], AL
skip:
INC SI
LOOP convert_loop

done:
; display converted string
LEA DX, BUFFER + 2
MOV AH, 09h
INT 21h

; exit
MOV AH, 4Ch
INT 21h
main ENDP
END main

```

Explanation: just subtract 20h or 32 from every character using loop



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emulator screen (80x25 chars)
Enter letters in lowercase: hello ahmad
uppercase: HELLO AHMAD

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